AVISHKAR MAHESH PAWAR

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Professional Summary

Computer Science student with hands-on experience building a complete database system from scratch (MiniDB, all 6 phases). Implemented production-grade storage engine with B+ Tree indexing, MVCC transaction isolation, full SQL-like query processing with JOINs, Raft consensus, and sharding. Strong understanding of database internals including buffer pool management, write-ahead logging, and crash recovery. GATE 2026 qualified (AIR 3124) with excellent academic record.

Education

MIT Academy of Engineering, Pune 2022 - 2026
B.Tech. - Computer Engineering — CGPA: 9.24/10

Projects

MiniDB — Distributed Database with ACID Transactions

C++17, Boost.Asio, Protocol Buffers, CMake

- Developed production-grade storage engine with B+ Tree indexing supporting range queries and point lookups with O(log n) complexity
- Implemented full query processing pipeline with SQL-like parser and executor supporting SELECT, INSERT, UPDATE, DELETE with JOIN operations across distributed tables
- Built MVCC (Multi-Version Concurrency Control) for snapshot isolation, enabling 1000+ concurrent transactions without read locks, with cost-based query optimization

AI-Powered Vector Database

Python 3.11, FastAPI, NumPy, Numba

- Developed specialized storage engine for high-dimensional vector data with efficient serialization and compression
- Implemented similarity search with configurable distance metrics (cosine, euclidean, dot product) and filtering capabilities
- Built metadata indexing system enabling hybrid search combining vector similarity and structured queries

Event Streaming Platform

Java 17, Spring Boot, Netty, Apache Kafka Protocol

- Developed REST API using Spring Boot for cluster management, topic creation, and real-time monitoring with Prometheus integration
- Optimized producer batching and compression (Snappy/LZ4) reducing network bandwidth by 70%
- Containerized entire platform using Docker with multi-node deployment configuration

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript
Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js
AI/ML: TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba
Databases: PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations
Tools: Git, Docker, Linux, AWS, CMake, Maven, Prometheus
Distributed Systems: Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing
CS Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

Experience

OSMOS Jul 2025 - Present

Software Developer

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

Core-Decimal Solutions Feb 2025 - Aug 2025

Software Developer

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

Achievements & Certifications

GATE 2026: Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)

GATE 2025: Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100

First Place in Technodium 25: Won first place in technical competition

Certifications: CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation

Extra-Curricular Activities

Outdoor Sports & Games: Active participant in outdoor sports and recreational activities